

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of B. Pharm Examination

University Theory Examination Dec- 2016

MA121 Basic Mathematics

Date: 28/12/2016, Wednesday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION I is 30 minutes. Answers to SECTION I must be written during the first 30 minutes of the examination.
3. Answers to SECTION II and SECTION III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION I

Q.1. Darken the circle of the correct/most suitable choice. [20]

1. Let A, B and C are three sets, then which of the following set relation is true?

(A) $A \cup (B \cap C) = (A \cap B) \cup (A \cap C)$ (C) $(A \cap B \cap C)' = (A' \cap B' \cap C')$
(B) $A \cap (B \cup C) = (A \cap B) \cap (A \cap C)$ (D) $(A \cap B \cap C)' = (A' \cup B' \cup C')$

2. If $A = \{1, 3\}$ and $B = \{3, 4, 5\}, C = \{4, 5, 6, 7\}$ then $A \cap B =$

(A) $\{1, 3\}$ (B) $\{3, 4\}$ (C) $\{3\}$ (D) $\{1, 3, 4, 5\}$

3. Suppose $n(U) = 70, n(A) = 30, n(B) = 45$ and $n(A \cup B) = 65$ then

$n(A' \cap B') =$ (A) 5 (B) 25 (C) 10 (D) 20

4. Consider the sets $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}, C = \{r, s, t, u\}$ and functions $f_1 = \{(a, 4), (b, 1), (c, 2)\}, f_2 = \{(1, r), (2, t), (3, u), (4, s)\}$ then

(A) f_1 is not one-one but onto (C) f_2 is one-one and onto
(B) f_2 is one-one but not onto (D) f_1 is one-one and onto

5. If $y = \log_b x$ then (A) $b^y = x$ (B) $x^y = b$ (C) $b^x = y$ (D) $y^b = x$

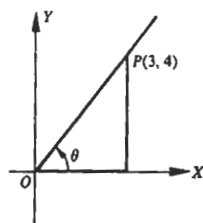
6. $\lim_{x \rightarrow c} f(x) = 5$ and $\lim_{x \rightarrow c} g(x) = -2$, then $\lim_{x \rightarrow c} \frac{g(x)}{(f(x) - 1)} =$

(A) $\frac{1}{2}$ (B) 2 (C) $-\frac{1}{2}$ (D) -2

7. A function $f(x)$ is continuous at an interior point $x = c$ of its domain if and only if

- (A) $f(c)$ exists
 (B) $\lim_{x \rightarrow c} f(x)$ exists
 (C) $\lim_{x \rightarrow c} f(x) = f(c)$
 (D) All three conditions

8. If $P(3, 4)$ is point on the terminal side of angle θ then $\sin \theta =$



- (A) $\frac{3}{5}$ (B) $\frac{4}{5}$ (C) $\frac{3}{4}$ (D) $\frac{5}{4}$

9. Which of the following relationship is true?

- (A) $\tan \theta = \frac{1}{\pm \sqrt{1 - \sin^2 \theta}}$
 (B) $\tan \theta = \frac{\pm \sqrt{1 - \sin^2 \theta}}{\sin \theta}$
 (C) $\tan \theta = \cos^2 \theta - 1$
 (D) $\tan \theta = \frac{\sin \theta}{\pm \sqrt{1 - \sin^2 \theta}}$

10. The slope of the curve $y = \frac{1}{1-x}$ at point $x = 3$ is

- (A) $-\frac{1}{4}$ (B) $-\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{2}$

11. If u and v are differentiable functions of x , then $\frac{d}{dx}(u + v) =$

- (A) $\frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$ (B) $\frac{du}{dx} + \frac{dv}{dx}$ (C) $v \frac{du}{dx} + u \frac{dv}{dx}$ (D) $v \frac{du}{dx} - u \frac{dv}{dx}$

12. The second derivative of the function $y = -x^2 + 3$ is

- (A) $-2x + 3$ (B) 3 (C) -2 (D) 0

13. An anti-derivative of the function $f(x) = \frac{1}{\sqrt{x}}$ is

- (A) $2\sqrt{x} + C$ (B) $2\sqrt{x}$ (C) $\sqrt{x} + C$ (D) $\sqrt{x}$

14. Which of the following formula for integral $\int x \sin x dx$, is true?

- (A) $\frac{x^2}{2} \sin x + C$ (C) $-x \cos x + \sin x + C$
 (B) $-x \cos x + C$ (D) All three are true

15. If $A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 & 1 \\ 3 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ then

- (A) $|A| = |B|$ (B) $|A| = -|B|$ (C) $|A| \neq |B|$ (D) $|A| = |B| = 0$

16. If $F = \begin{bmatrix} 3 & 4 & -6 \\ 1 & -1 & k \\ a & 1 & 1 \end{bmatrix}$, then cofactor of element k is

- (A) $4a - 3$ (B) $k(4a - 3)$ (C) $-4a + 3$ (D) $-k(4a - 3)$

17. The inverse of matrix $\begin{bmatrix} -1/3 & 0 \\ 0 & 5 \end{bmatrix}$ is

- (A) $\begin{bmatrix} 3 & 0 \\ 0 & -1/5 \end{bmatrix}$ (B) $\begin{bmatrix} -3 & 0 \\ 0 & -1/5 \end{bmatrix}$ (C) $\begin{bmatrix} 3 & 0 \\ 0 & 1/5 \end{bmatrix}$ (D) $\begin{bmatrix} -3 & 0 \\ 0 & 1/5 \end{bmatrix}$

18. Which of the following matrices is singular?

- (A) $\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 2 \\ 5 & 8 \end{bmatrix}$ (C) $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} -1 & 6 \\ 3 & 9 \end{bmatrix}$

19. The quadratic equation $ax^2 + bx + c = 0, a \neq 0$ has identicle roots if

- (A) $\sqrt{b^2 - 4ac} < 0$ (B) $\sqrt{b^2 - 4ac} = 0$ (C) $\sqrt{b^2 - 4ac} > 0$ (D) $\sqrt{b^2 - 4ac} \neq 0$

20. Suppose we have to solve system of two equations

$a_1x + b_1y = c_1, a_2x + b_2y = c_2$, then x and y can be determined uniquely if

- (A) $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = 0$ (B) $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \neq 0$ (C) $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = 1$ (D) $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = -1$

SECTION II

Q.2. Attempt any **Two** of the following. [10]

A Construct the venn diagram of the set relations [05]

(a) $(A \cup B) \cap C'$ (b) $A \cap B \cap C'$ (c) $(A \cap B') \cup (A' \cap C)$ (d) $A \cap B \cap C$

B Consider the Universal set $S = \{1, 2, \dots, 9\}$ and the sets

$A = \{1, 2, 3, 4\}, B = \{4, 5, 6, 7\}, C = \{5, 6, 7, 8, 9\}, D = \{1, 3, 5, 7, 9\}$ Find [05]

(a) $A \cup B$ (b) $B \cup D$ (c) $A \cap (B \cap D)$ (d) $D \cup C'$

C 1. Sketch the graph of $f(x) = x^2 + x - 6$ [03]

2. Let $A = \{1, 2\}, B = \{a, b, c\}$ and $C = \{c, d\}$ determine [02]

(a) $(A \times C) \cup (A \times B)$ (b) $A \times (B \cap C)$

Q.3. Attempt any **Two** of the following. [10]

A Evaluate the limit $\lim_{x \rightarrow 4} \frac{4-x}{5-\sqrt{x^2+9}}$ [05]

B Verify the identity $\frac{\tan x - \sin x}{\sin^3 x} = \frac{\sec x}{1 + \cos x}$ [05]

C At what points is the functions $y = \frac{1}{x-2} - 3x$ continuous? [05]

SECTION III

Q.4. Attempt any **Four** of the following. [20]

A Find the first and second derivatives of the function $y = \frac{4x^3}{3} - x$ [05]

B Consider the matrices $A = \begin{bmatrix} 0 & 5 & 2 \\ 5 & 1 & 4 \\ 2 & 4 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 & 4 \\ -3 & 1 & -7 \\ -4 & 7 & 2 \end{bmatrix}$ [05]

Find matrix product

(a) AB (b) BA

C Find the derivative of $y = \frac{\cos x}{1 - \sin x}$ [05]

D If $A = \begin{bmatrix} 2 & 3 & 1 \\ 6 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ Then verify $(AB)' = (B'A')$ [05]

E Find the derivative of $y = \frac{x^2 - 1}{x^2 + 1}$ [05]

F If $A = \begin{bmatrix} 2 & 3 \\ 1 & -4 \\ 1 & 0 \end{bmatrix}, B = \begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 1 & -4 \end{bmatrix}$ and $C = \begin{bmatrix} 2 & 1 \\ 2 & 0 \\ 0 & 1 \end{bmatrix}$ Find [05]

(a) $2A + B - 4C$ (b) $(A + B)'$

Q.5. Attempt any **Four** of the following. [20]

A Evaluate integral $\int_1^4 \sqrt{4x-1} dx$ [05]

B Find inverse of matrix A where $A = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 4 & 0 \\ 6 & 2 & 5 \end{bmatrix}$ [05]

C Evaluate the integral $\int (2x+1)^3 dx$ [05]

D If $A = \begin{bmatrix} 2 & 3 & 1 \\ 6 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ Then verify $|AB| = |A||B|$ [05]

E Evaluate the integral $\int_1^4 \left(\frac{3}{2}\sqrt{x} - \frac{4}{x^2} \right) dx$ [05]

F 1. Solve the quadratic equation $3x^2 - 2x - 4 = 0$ [02]

2. Solve using Cramer's rule $x - 2y = 1, 2x - 3y = 3$ [03]
